Peng Li

Stockholm, Sweden | +46 79 010 31 96
pengli.2810@gmail.com

6 November 2026

William Smati and the Recruitment Team
CorPower Ocean
Stockholm, Sweden

Application: Hydrodynamics Analysis Engineer – Uncrewed Surface Vessel

Dear William Smati and Recruitment Team,

I am applying for the Hydrodynamics Analysis Engineer position at CorPower Ocean. I bring experience in computational fluid dynamics (CFD), fluid–structure interaction (FSI), and structural dynamics, together with industrial model development and validation. I would like to apply these strengths to understanding vessel response and supporting design decisions for your uncrewed installation vessel. I am currently completing my Ph.D. research at KTH Royal Institute of Technology, with my defence scheduled for 4 December 2026.

My doctoral work in upper-airway aeroacoustics predicts how flexible structures respond to unsteady fluid loading. I developed three-dimensional STAR-CCM+ models coupling compressible LES with FEM-based structural mechanics, using strong two-way FSI, hyperelastic materials, and moving overset meshes. By comparing flow conditions, material stiffness, and geometry variants, I investigate how loads and structural properties affect deformation and vibration. Modal analysis, cross-spectra, and force–velocity phase relationships help me explain the mechanisms behind the predicted response. These are transferable skills for analysing coupled fluid–mechanical systems and comparing engineering design alternatives.

At Joyson Safety Systems, I developed Python-based transient models of pyrotechnical initiators and coupled their outputs to OpenFOAM simulations of shock-wave propagation in airbag inflators. I compared predicted pressure histories with measurements and used parameter-sensitivity studies to assess the influence of component characteristics on performance and structural integrity. This experience taught me to connect simulation results with physical behaviour and practical product-development questions.

Verification is central to my modelling approach. I conduct mesh and time-step convergence studies, compare analytical predictions with finite-element calculations, and use NAISS HPC resources at PDC and NSC for demanding simulations. My additional research includes analytical plate bending and buckling, COMSOL modelling of low-Mach-number flows, and validation of reduced combustion models. A master's degree in Mechanics from TU Darmstadt, with a final grade of 1.33, provides a foundation in fluid and structural mechanics and numerical methods for partial differential equations.

CorPower's combination of modelling, vessel development, and feedback from offshore operations appeals to me because it connects analysis directly to engineering decisions. My current simulation tools are STAR-CCM+, OpenFOAM, COMSOL, and ABAQUS; I would build proficiency in OrcaFlex and marine installation analysis as I extend my CFD–FSI background into this field. I would contribute scientific programming, systematic model verification, and experience collaborating across fluid mechanics, structural dynamics, and acoustics. I use English for scientific writing and research communication, and have led analysis and manuscript preparation for peer-reviewed publications.

I would welcome the opportunity to discuss how my experience in coupled dynamics, industrial simulation, and numerical model development could contribute to your Marine Operations team and the deployment of wave-energy technology.

Kind regards,

Peng Li